

Detim Zhao

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Education

Bachelor of Science, Software Engineering – May 2026

- Arizona State University, Polytechnic, AZ – GPA: 4.0/4.0
 - Barrett (Honors), Dean's List (Fall 2022, Spring 2023)

Skills

- Programming – Java, Python, C, PostgreSQL, SQL
- Tools – Git, GitHub, LaTeX, Photoshop, Illustrator, Gazebo, Robot Operating System

Work Experience

Undergraduate Researcher – Texas A&M-Corpus Christi / Corpus Christi, TX 6/5/23-8/11/23

- Researched, innovated, and developed an intuitive method of interaction between a human and a swarm of drones.
- Developed and contributed to a Python program to spawn, connect, and operate multiple PX4 drones in a gazebo simulation through terminal commands **(Python)**
 - Additional software: *Mavsdk, PX4 Autopilot, Gazebo, ROS*
- Employed preprocessing techniques on two datasets and conducted training and testing on gesture recognition for five machine learning models (KNN, SVM, Random Forest, Naive Bayes, and Ensemble) on both datasets. **(Python)**
 - Python libraries: *pandas, scikit-learn, matplotlib, seaborn*
- Co-authored a research paper for publication that documents findings and work **(LaTeX)**

Projects

Gesture Recognition of Hand Gestures with ML

- Streamlined and performed data preprocessing tasks (data cleaning and handling missing values) on both small and large datasets, ensuring data consistency and quality. **(Python)**
- Designed, trained, and evaluated KNN, SVM, Random Forest, Naive Bayes, and Ensemble machine learning models with both small and large datasets. Improved model accuracy by ~10% for some models through hyperparameter tuning. **(Python, scikit-learn)**

Control Drone (PX4) Swarm in Gazebo

- Automated script to spawn and connect to multiple PX4 drones in a gazebo simulation. **(Python)**
- Designed options to select a specific world and specify the number of drones. **(Python)**
- Implemented manual input commands from the terminal to control individual drones or groups of drones. **(Python)**

Console-Based Chat Application

- Designed console-based directions and options for users to register, log in, update info, create a chat room, join a chat room, and log out. **(Java)**
- Implemented commands when the user is in a chat room to display active users, chat history, and more. Organized information about the user and messages are stored and managed through the local host pSQL database. **(Java, PostgreSQL)**

Maze Generator and Solver

- Developed a GUI-based maze generator and solver where the user specifies the dimensions of the maze. **(Java, JavaFX)**
- Implemented a recursive backtracking algorithm to solve the maze and draw the solution path from the start to the end point of the maze. **(Java, JavaFX)**